



Animal Movement Analyses for Ecologists



Smithsonian Conservation
Biology Institute



SENCKENBERG
NATURFORSCHUNG



@ Fudan University | August 4 - 10, 2025

Welcome!



- Sponsors, Instructors, and Participants Introductions
- Course overviews and schedules
- Questions/Comments

Please introduce yourself

Name

Affiliation

What are you looking for from this course



MOVEBANK

animal tracking

Scientists have been systematically collecting animal movement data since the 1800s, employing ever-evolving technologies:

Fred Zwicky



Data collection methods:



1800s
Bird banding

1940s
Radar



Patricia Lim



1960s
Radio-telemetry

1990s
GPS technology

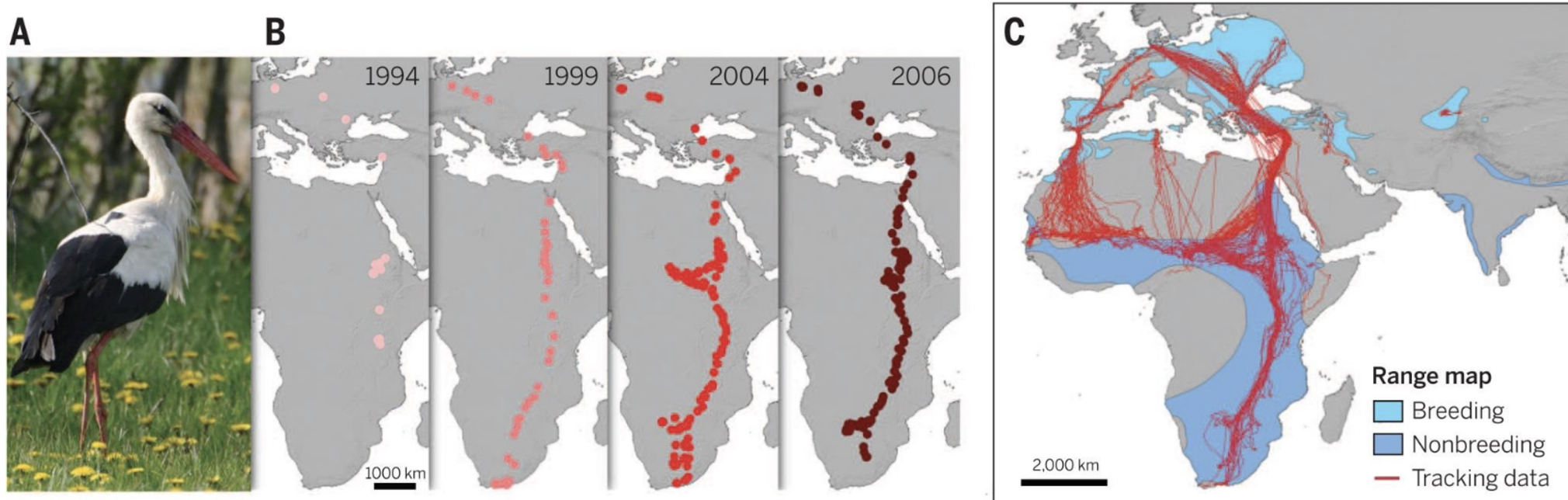


2000s
Nanotags,
drones

Biologging technologies
(animal-carried devices that
collect and/or transmit data)

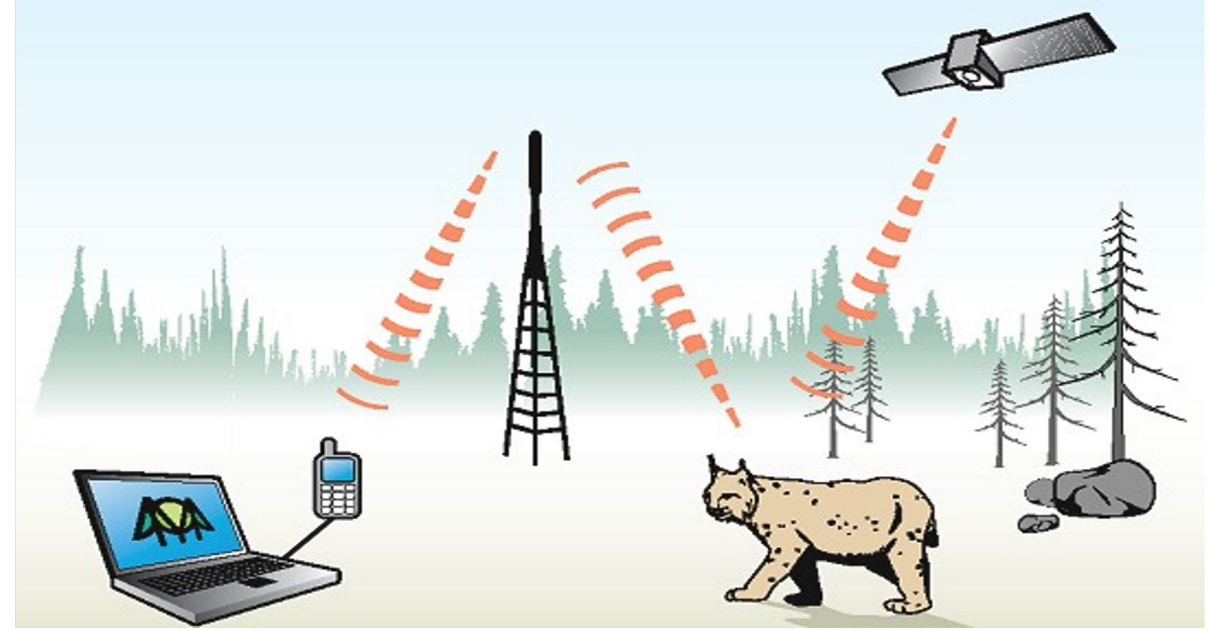
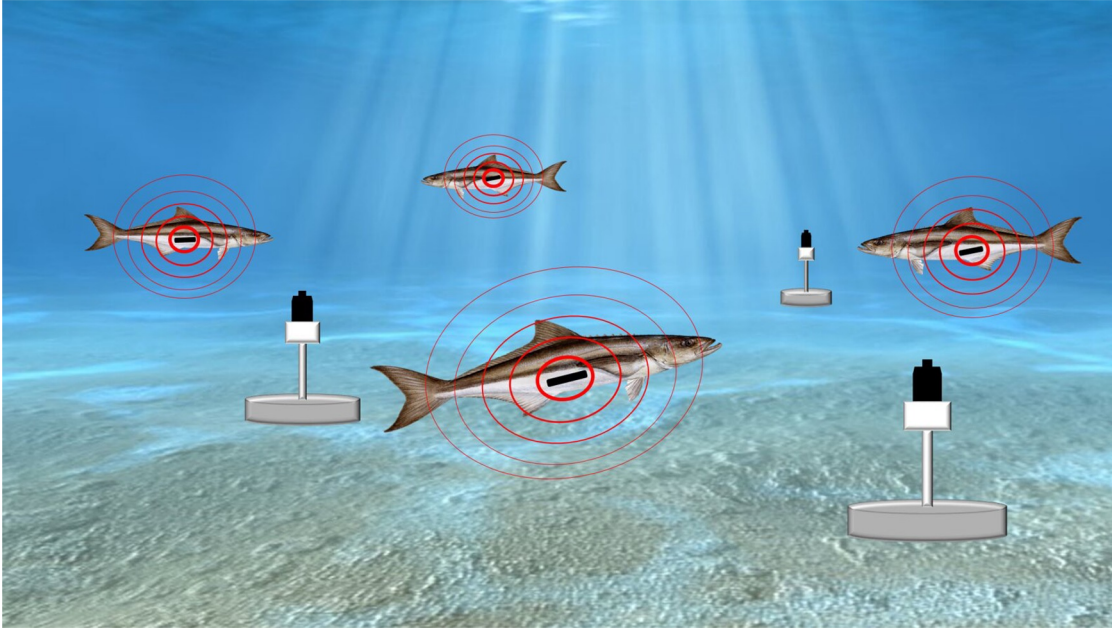
animal tracking

- Adapted from Kays et al. (2015)
 - A white stork (*Ciconia ciconia*) (A) was tagged with a GPS tracking device as a 3-year-old nonreproductive juvenile in Germany in 1994, and was tracked until her death in 2006 (B).



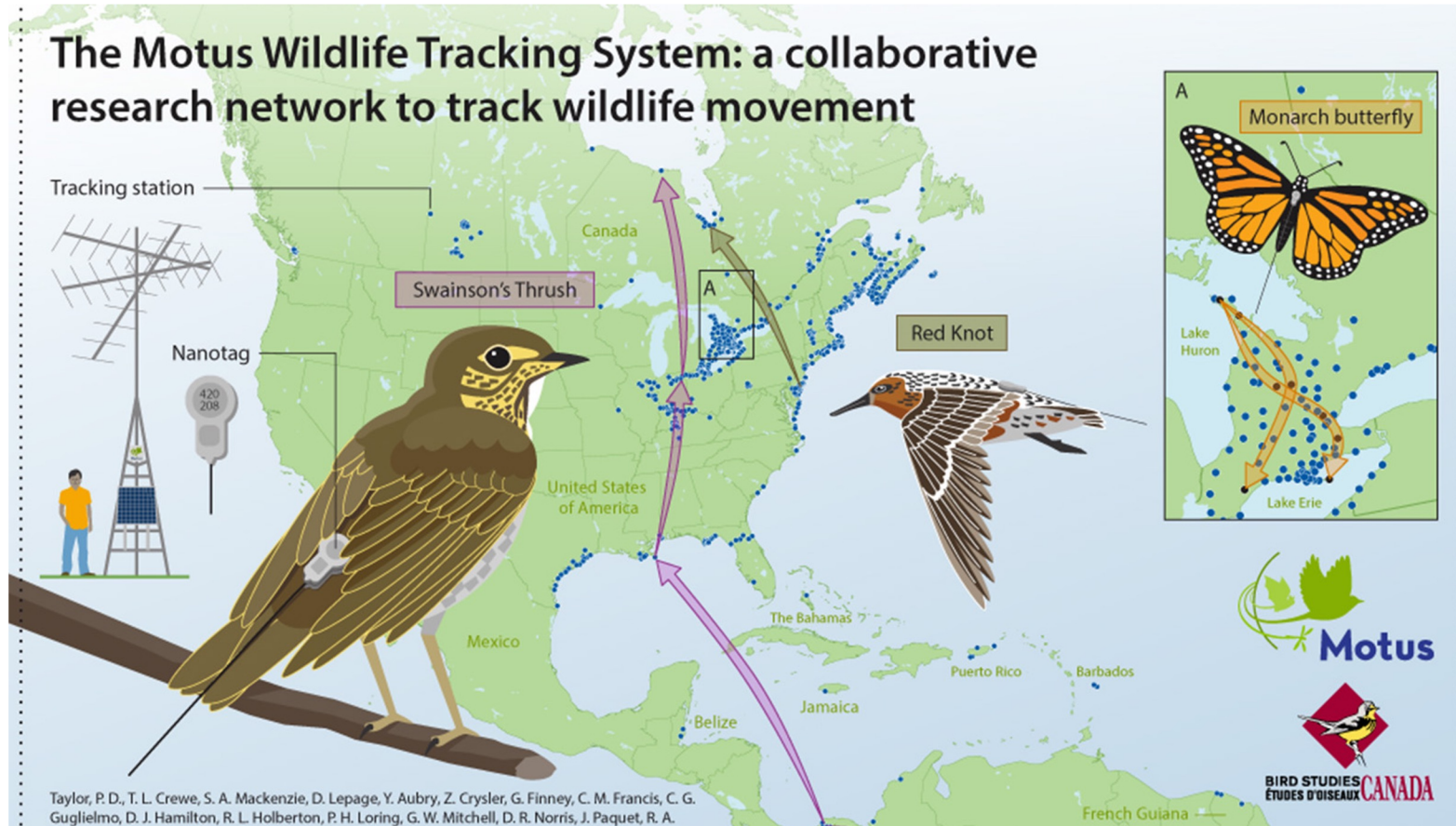
animal tracking

Eulerian vs. Lagrangian



animal tracking

Eulerian + Lagrangian



why track animals

Animal movement is fundamental for life on Earth

For natural system

- Pest control
- Seed dispersal
- Pollination
- Foraging
- Nutrient cycling
- ...

For human societies

- Food sources
- Disease transmission
- Safety
- Tourism
- Cultural and spiritual entities
- Environmental sentinel
- ...



why track animals

Animal movements are building blocks of ecology across scales



Individual level

- ecophysiology
- life history

why track animals

Animal movements are building blocks of ecology across scales



Individual level

- ecophysiology
- life history



Population level

- collective behavior
- social behavior

why track animals

Animal movements are building blocks of ecology across scales



Individual level

- ecophysiology
- life history



Population level

- collective behavior
- social behavior



Community level

- trophic interaction
- disease ecology

why track animals

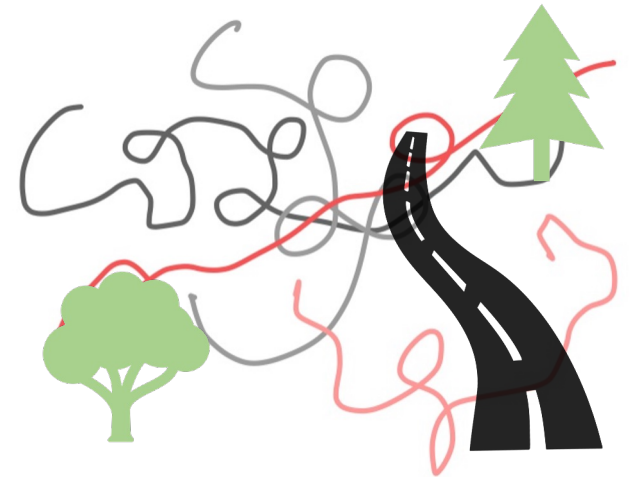
Animal movements are building blocks of ecology across scales



Individual level



Population level



Community level



Interaction with the natural or human-mediated environment

why track animals

Increasing data columns and resolutions allowing movement data to address increasingly complicated biological, ecological, and conservation questions

nature ecology & evolution

Perspective

<https://doi.org/10.1038/s41559-024-02585-4>

Establishing bio-logging data collections as dynamic archives of animal life on Earth

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Check for updates

Sarah C. Davidson^{1,2,3,22}✉, Francesca Cagnacci^{4,5,22}✉, Peggy Newman⁶, Holger Dettki^{7,8}, Ferdinando Urbano⁹, Peter Desmet¹⁰, Lenore Bajona^{11,12}, Edmund Bryant^{13,14}, Ana P. B. Carneiro¹⁵, Maria P. Dias¹⁶, El Fujioka¹⁷, David Gambin¹⁸, Xavier Hoenner¹⁹, Colin Hunter¹³, Akiko Kato²⁰, Connie Y. Kot^{17,21}, Bart Kranstauber²², Chi Hin Lam^{23,24}, Denis Lepage²⁵, Hemal Naik^{26,27}, Jonathan D. Pye¹¹, Ana M. M. Sequeira^{28,29}, Vardis M. Tsontos³⁰, Emiel van Loon²², Danny Vo¹³ & Christian Rutz³¹✉

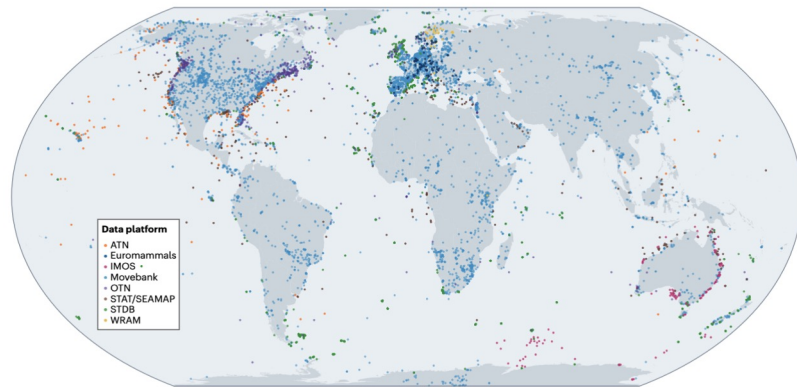


Fig. 1 Map of discoverable bio-logging studies stored on selected community bio-logging data platforms. Each point represents a dataset with distinct methods, funding and owners as of August–September 2024; this probably represents a small fraction of bio-logging studies undertaken³¹. Of 9,647 studies shown, only 1,203 (12%) allow public download (Supplementary Table 1). Data platforms are Euro mammals (Euroboar and Eurodeer networks,

<https://euromammals.org>), IMOS Australian Ocean Data Network (<https://portal.aodn.org.au>), Movebank (<https://movebank.org>), OTN (<https://oceantrackingnetwork.org>), Satellite Tracking and Analysis Tool datasets shared in OBIS-SEAMAP (STAT/SEAMAP, <https://seamap.env.duke.edu/partner/STAT>), Seabird Tracking Database (STDB, <https://data.seabird-tracking.org>), US ATN (<https://atn.joos.us>) and WRAM (<https://wram.slu.se>).

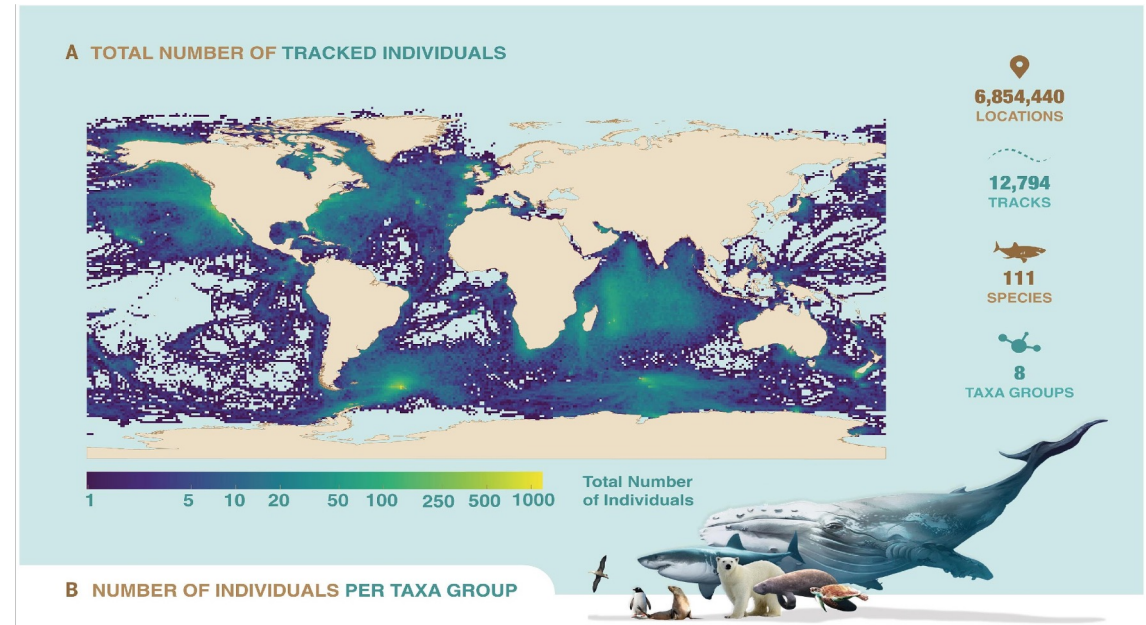
RESEARCH ARTICLE | MARINE CONSERVATION

f X t in d w s e

Global tracking of marine megafauna space use reveals how to achieve conservation targets

ANA M. M. SEQUEIRA¹✉, JORGE P. RODRÍGUEZ²✉, SARAH A. MARLEY³✉, HANNAH J. CALICH⁴✉, MIRJAM VAN DER MHEEN⁵✉, MICHELLE VANCOMPERNOLLE⁶✉

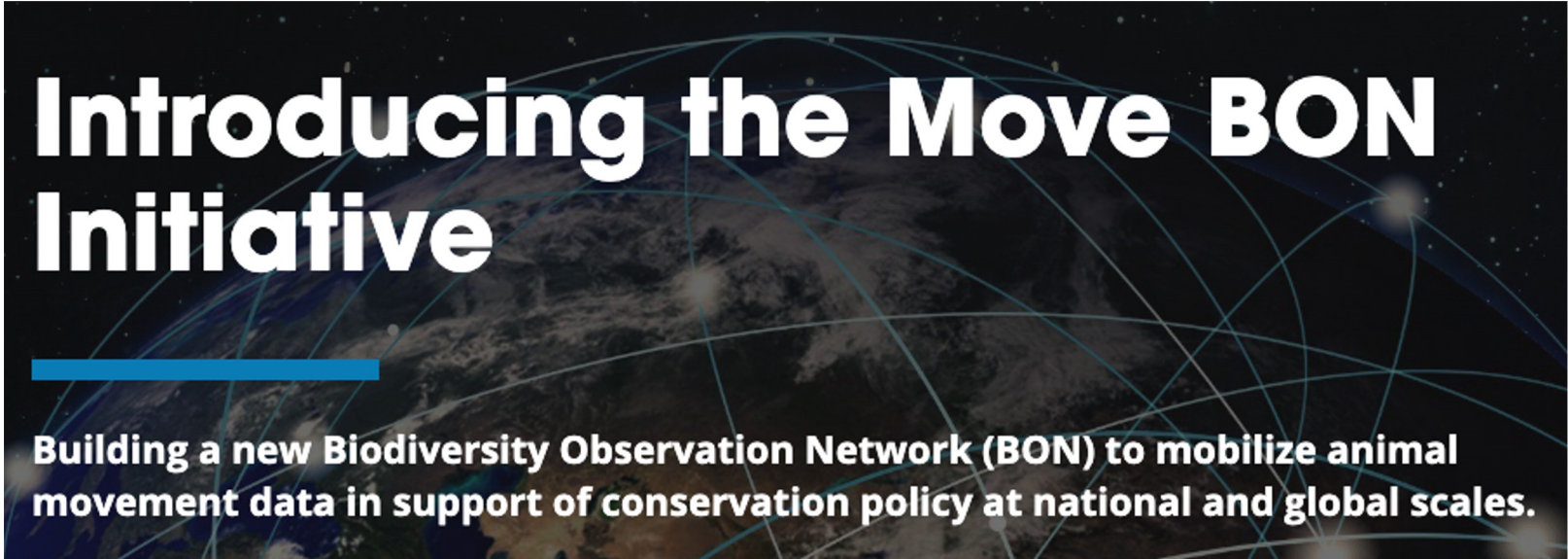
LUCY M. ARROWSMITH⁷✉, LAUREN R. PEEL⁸✉, NUNO QUEIROZ⁹✉, [...] AND VÍCTOR M. EGUÍLUZ¹⁰✉ +367 authors [Authors Info & Affiliations](#)



why track animals

Increasing data columns and resolutions allowing movement data to address increasingly complicated biological, ecological, and conservation questions

... and inform policy



Introducing the Move BON Initiative

Building a new Biodiversity Observation Network (BON) to mobilize animal movement data in support of conservation policy at national and global scales.

Aug 4

PM

Introduction to animal movement data
data import, cleaning, and visualization

Aug 5

AM

Movement metrics
building a discrete-time trajectory, mapping, and track-level movement metrics

PM

Movement metrics (cont.)
Metrics applications (Barrier Behavior Analysis, k-means clustering)

Aug 6

AM

Model-based behavioral insights
introduction to remote sensing and data annotation with environmental information, space use metrics with discrete methods

PM

Class field trip

Aug 7

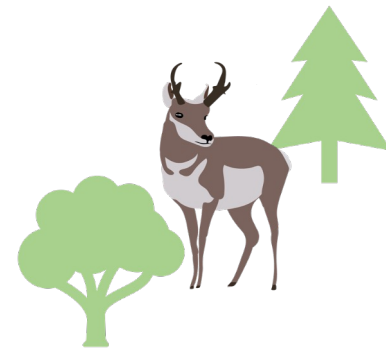
AM

Exercise
Introduction to Hidden Markov Models (HMM);

Course schedule Pt 1
Wenjing

Jared +

Intro to movement data and discrete methods



Aug 7

PM

Introduction to continuous-time movement modeling (ctmm)
Review of key concepts, ctmm introduction, autocorrelation

Aug 8

AM

Home-range estimation, Population-level meta-analysis
AKDE, population-range estimation

PM

Continuous-time movement modeling and applications
Occurrence distributions, speed & distance estimation, location error, interactions

Aug 9

AM

Habitat suitability analyses
Resource selection functions, study design, period patterns of space use, species distribution modeling

PM

Integrative step selection function
SSFs

Aug 10

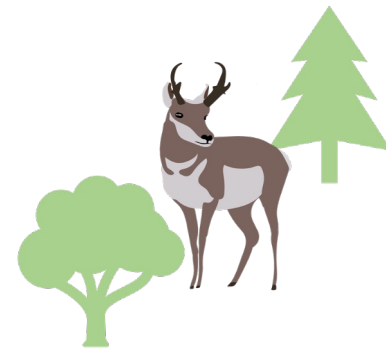
AM

Individual work with instructors
Work on your own data

Course schedule Pt2
Erika

Chris +

Continuous methods of movement analysis



Datasets

White-bearded Wildebeest (*Connochaetes taurinus*)

Total individuals monitored: 36

Total populations: 3

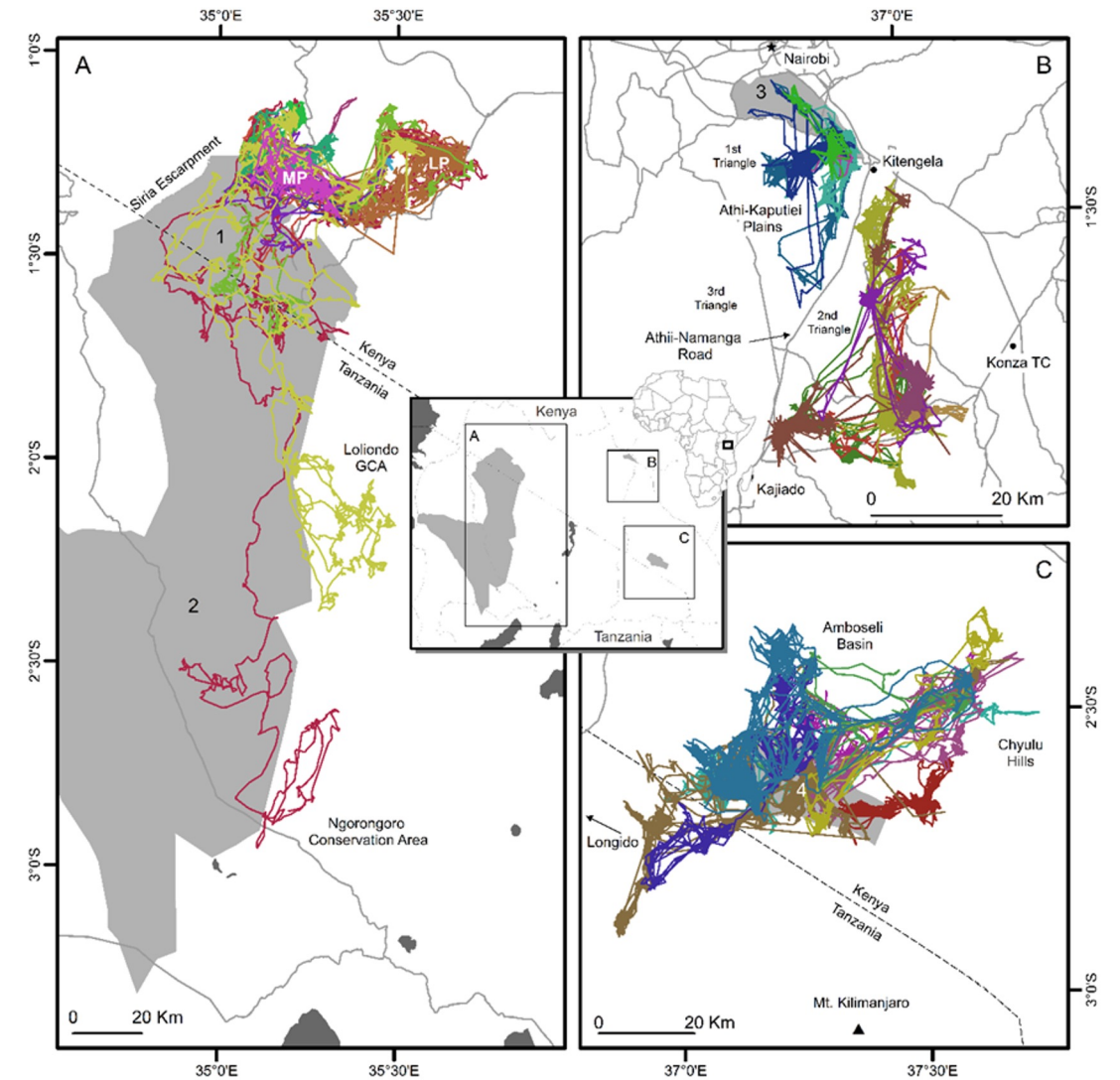
Amboseli Basin ($n = 9$)

Athi-Kaputiei Plains ($n = 12$)

Mara-Loita Plains ($n = 15$)

Collection date: 2010-2013

Total number of points: 276,706



Questions?